Monetary Policy as Decentralized Control: Near-Optimality of Quantized Policy Rules

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Abstract

This paper studies macroeconomic stabilization as a cooperative game: the central bank and private agents share the common goal of stabilizing the output around potential but under an information differential: policymakers observe private signals about changes in the natural rate of interest that households and firms either lack or find costly to monitor; however, interest rate adjustments are costly making it infeasible to offset changes in the natural rate completely. Private agents adjust inflation expectations based on noisy observations of the *output gap*. Importantly, central bank policy influences private sector's information by moving the output gap, which serves as an implicit communication channel. This creates a signaling incentive for policymakers: stabilization can be achieved by making it easier for private agents to decipher the output gap signal and stabilize via the expectations channel avoiding costly interest-rate adjustments. This paper draws an analogy to *Witsenhausen's counterexample*, a canonical problem in decentralized control theory, where optimal strategies can be non-linear. The analysis shows that quantized policy rules—such as adjusting interest rates in discrete steps—can outperform linear feedback rules like the Taylor rule.

1 Introduction

The stabilization role of monetary policy is inherently a problem of decentralized control. Policymakers determine the short-term nominal interest rate, while the evolution of the economy depends critically on how private agents form expectations about the future. Hence, optimal macroeconomic stabilization can be thought of as a cooperative *team decision problem* in which multiple actors influence aggregate outcomes. But a central bank and the private agents could possibly operate under distinct information sets or beliefs. The knowledge differential can bear on the conduct of monetary policy.

This paper studies optimal policy in a setting where the central bank and the private sector jointly aim to stabilize the economy against shocks, but under a *nonclassical* information structure. The central bank observes an economic disturbance, such as a change in the natural rate of interest - that is not directly observed by private agents. Policy adjustments are costly, reflecting quadratic costs for changes in the policy rate. Policy actions influence aggregate demand, which private agents only observe through a noisy channel. Agents can adjust expectations costlessly but do so under imperfect information: they are *blurry*.

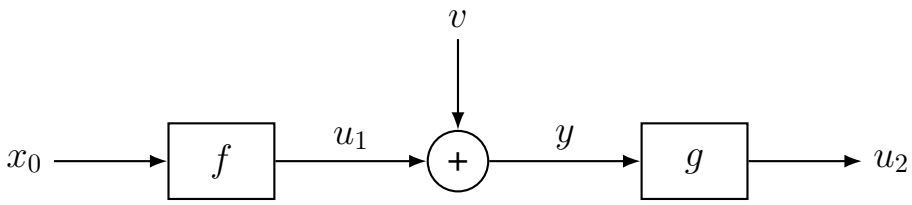
The key insight is that monetary policy serves a dual purpose: it both stabilizes the economy and communicates information about the underlying fundamental. In this sense, central bank policy actions affect the information structure faced by the private agents. The policy instrument therefore plays an informational as well as a stabilization role. The optimal policy must balance the incentive to smooth shocks against the value of conveying information to the private agents, who condition their expectations on the observed aggregate state.

This structure mirrors the *Witsenhausen's counterexample* (Witsenhausen (1968)), a canonical problem in decentralized stochastic control that showed that nonlinear control policies can be optimal even in Linear Quadratic Gaussian (LQG) setting in which two players must coordinate towards achieving a common goal, but under different information sets. One player's action can serve to communicate information to the other player. Hence, the optimal control problem embeds an optimal communication problem within it. This setup serves as an important fundamental building block to think about the signaling incentive of a central bank and its implications for monetary policy.

Though recent literature, such as Grover (2011) and Li, Marden and Shamma (2010) have studied the toy example and explored some extensions, the optimal solution to the problem is still unknown. The discrete-version of the problem is known to be NP-complete (Papadimitriou and Tsitsiklis (1985)), implying that finding the optimal solution is computationally infeasible. However, it has been shown that a certain class of

2 Optimal Control under a Nonclassical Information Structures

2.1 Witsenhausen's Counterexample



Consider a two-stage team control problem. Let the initial state be a random variable $x_0 \sim \mathcal{N}(0, \sigma_x^2)$. There are two controllers:

1. Player 1 observes x_0 and alters state to $u_1 = f_1(x_0)$ by adding a control input $u_1 - x_0$.
2. A noisy channel transmits u_1. Player 2 observes $y = u_1 + v$, where $v \sim \mathcal{N}(0, \sigma_v^2)$, and chooses $u_2 = g(y)$, adding an input $u_2 - u_1$.

The total cost is quadratic:

$$\mathcal{L} = \mathbb{E}\big[k^2(u_1 - x_0)^2 + u_2^2\big], \tag{1}$$

where $k > 0$ penalizes the first control effort and the second term penalizes the squared deviations of the final state u_2 from 0.

2.2 Key features

- Decentralized information: Player 2 does not know x_0 directly, only the noisy signal y
- Signaling incentive: Player 1's action u_1 both changes the system state and communicates information about x_0 to Controller 2 through y
- Nonclassical information structure: The decision of the first controller affects the information available to the second, breaking the separation between estimation and control, thereby making nonlinear control policies optimal.

Despite the linear-quadratic-Gaussian (LQG) setting, the optimal encoding $f_1(x_0)$ is *nonlinear*. Witsenhausen (1968) showed that under this non-classical information structure, certain "quantized" mappings, where policy is characterized by discrete "jumps" can outperform all linear strategies. The insight is that it can be optimal to use costly control actions for *communication*: small, discrete shifts in u_1 make y more informative to Player 2 who can then stabilize by making cost-less adjustment $u_2 - u_1$. Hence, the asymmetric information structure of the problem make it optimal to use the control signal as a communication channel.

In the next section, we write down a simple economic model in which a central bank and the private sector must optimally stabilize against fundamental shocks under a shared objective. There is a *knowledge gradient* between the two: the central bank can perfectly observe shocks to the fundamental, but must employ a costly control input. The private sector on the other hand, has access to a control input that is costless but must operate under uncertainty about the

fundamental, only being able to observe a noisy realization of the central bank's action. I show that this problem maps directly to the setup described earlier.

3 A Stylized Economy

To illustrate the main features of the problem, we may consider a simple economy with the output gap x_t to be given by the following IS-curve:

$$x_t = x_t^e - \tau(i_t - r_t^n - \pi_t^e) \tag{2}$$

where τ is the intertemporal elasticity of substitution, i_t is the nominal rate of interest set by the central bank, r_t^n is the natural rate of interest and π_t^e denotes the Private Sector's one-period ahead inflation expectations at time t. Hence, the output gap x_t is determined by the private agents' one-period ahead expectation of the output gap, given by x_t^e and the real rate gap, or the difference between the real rate of interest $i_t - \pi_t^e$ and the natural rate r_t^n.

Let CB denote the central bank and PS denote the private agents. In the rest of this section, I describe a team stabilization problem with two cooperative players CB and PS. They share a common goal, but do not share common information. The next subsection describes the information structure of the game.

3.1 Information structure

At the beginning of each period t, CB observes an *iid* output-gap disturbance $x_t^0 \sim N(0, \sigma_0^2)$:

Consistent with Equation 2, CB adjusts interest rates to i_t, resulting in an intermediate output gap x_t^1 given by:

$$x_t^1 = x_t^0 - \tau(i_t - i_{t-1}) \tag{3}$$

We may refer to the time-t interest rate adjustment $i_t - i_{t-1}$ as the central bank's time-t *action* and denote it by

$$a_t^{CB}(x_t^0) = -\tau(i_t - i_{t-1})$$

Then, the intermediate output gap is just the sum of the initial disturbance and the CB's action:

$$x_t^1 = x_t^0 + a_t^{CB}$$

PS observes a noisy signal ξ_t about x_t^1 confounded with Gaussian noise, given by:

$$\xi_t = x_t^1 + e_t, \quad e_t \sim N(0, \sigma_e^2) \tag{4}$$

Conditional on the observed ξ_t, PS updates its expectations, choosing π_t^e and x_t^e. We can define a reduced-form construct called PS action as

$$a_t^{PS}(\xi_t) = x_t^e - x_{t-1}^e - \tau(\pi_t^e - \pi_{t-1}^e)$$

Finally, as a result of the initial output gap disturbance x_t^0, CB action a_t^{CB} and PS action a_t^{PS}, period-t output gap x_t realizes

$$x_t = x_t^1 + a_t^{PS} \tag{5}$$

The social loss function is given by:

$$\mathcal{L} = x_t^2 + \lambda(i_t - i_{t-1})^2 \tag{6}$$

where the first term imposes a quadratic penalty on output gap second term penalizes changes in the nominal interest rates

We can think of the problem as a strategic game between the central bank and the private sector. The central bank chooses an action $a_t^{CB}(x_t^0)$ as a function of the initial disturbance x_t^0 so as to set the intermediate output gap according to some functional mapping $x_t^1 = f(x_t^0)$. The private sector observes the signal ξ_t and chooses its action a_t^{PS} according to a functional mapping $a_t^{PS} = g(\xi_t)$. Both players may coordinate and choose f and g such that the expected loss given by Equation 6 is minimized.

3.2 Equivalence to a two-stage nonclassical team problem

Minimizing $\mathbb{E}[L]$ over measurable policies (f, g) is equivalent (up to a positive constant normalization) to minimizing over measurable policies (f, g) the social objective

$$\mathcal{L}(f, g) \;=\; \mathbb{E}\Big[\,(x_t^1 + a_t^{PS})^2 \;+\; \gamma\,(a_t^{CB})^2\,\Big], \qquad \gamma \equiv \frac{\lambda}{\tau^2}, \tag{7}$$

subject to the non-classical information structure

$$x_t^0 \text{ observed by CB}, \quad x_t^1 = f(x_t^0) \tag{8}$$

$$\xi_t = x_t^1 + e_t \qquad e_t \sim N(0, \sigma_e^2)$$

$$\xi_t \text{ observed by PS}, \quad a_t^{PS}(\xi_t) = g(\xi_t). \tag{9}$$

4 Optimal Policy under Knowledge Gradient

This section shows, in a transparent calibration, that a simple *1-bit quantized policy rule* - a rule that maps the central bank's observed shock into only two possible policy stances; can strictly outperform the best *linear* (Taylor-style) rule when the private sector must infer the policy-relevant state from a noisy aggregate signal. The mechanism is the familiar one from nonclassical decentralized control: the central bank's action is simultaneously (i) a stabilization instrument and (ii) an informational signal that shapes private expectations.

4.1 Optimal Private Sector Response

For any fixed policy rule f by the central bank, the private sector's best response under quadratic loss is the conditional expectation:

$$g^\star(\xi_t) = \arg\min_g \mathbb{E}\big[(x_t^1 + g(\xi_t))^2\big] \;=\; -\mathbb{E}[x_t^1 \mid \xi].$$

Hence, the private sector's problem is to optimally "decode" the intermediate output gap x_t^1 given the noisy signal ξ_t.

For a given policy rule f, the induced loss conditional on optimal inference by the private sector is given by:

$$\mathcal{L}(f) = \min_g \mathcal{L}(f, g) = \mathbb{E}\big[\mathrm{Var}(x_t^1 \mid \xi)\big] + \gamma\, \mathbb{E}[(f(x_t^0) - x_t^0)^2] \tag{10}$$

The first term is the residual stabilization loss after optimal inference. This term represents the loss resulting from the private sector not being able to infer x_t^1 perfectly from the observation ξ_t. The second term represents the loss resulting from costly central bank stabilization. Policies that make x_1 *more easily inferable* from ξ reduce $\mathbb{E}[\mathrm{Var}(x_1 \mid \xi)]$, but may increase $\mathbb{E}[(x_1 - x_0)^2]$.

In order to illustrate how the non-classical information structure affects the nature of optimal policy, we compare a benchmark "linear" policy rule, where f is a linear (possibly affine) function with a "nonlinear" discrete policy rule where the central bank chooses x_t^1 from among two discrete levels.

4.2 Two limit cases

Under the optimal choice of the mappings f, g, the minimum loss $\mathcal{L}$ is minimized. Hence, the minimum loss achieved must be bounded above by the simple policy rules corresponding to two limit cases:

- Limit case 1: Central bank takes no action so that $x_t^1 = x_t^0$
- Limit case 2: Central bank stabilizes fully by taking costly action, without relying on the private sector action

Limit case 1: Passive CB In this case $a_t^{CB} = 0 \quad \forall \quad x_t^0$, implying $x_t^1 = x_t^0$.

Under this policy, PS observes $\xi_t = x_t^0 + e_t$, i.e a noisy signal about the initial disturbance Optimal action $g(\xi) = -\xi$. In this limit case, there is no control loss incurred in the first stage by central bank stabilization. $\mathcal{L} = \mathrm{V}ar(e_t) = \sigma_e^2$.

Limit case 2: CB stabilizes fully In this case, the central bank takes costly action in order to set $x_t^1 = 0$ without relying on private sector's action. This involves action $a_t^{CB}(x_t^0) = -x_t^0$.

Since CB fully stabilizes the shock in this case, x_t^1 is known to be 0. ξ_t is irrelevant and PS takes no action so that $a_t^P(\xi) = 0 \quad \forall \quad \xi$.

The loss in this case is entirely from the central bank action: $\gamma\sigma_0^2$

Thus, the loss is bounded above by $\min\{\sigma_e^2, \gamma\sigma_0^2\}$.

4.3 Linear benchmark

In this subsection, I solve the optimal policy problem restricting f to the space of linear functions in x_0

$$\mathcal{F}_{\text{lin}} = \{f(x_t^0) = -\alpha x_t^0 : \ \alpha \in \mathbb{R}\}.$$

Under a linear rule $x_1 = \alpha x_0$, the problem is Gaussian. The optimal decoder is also a linear function of the observed ξ_t,

$$a^{*PS}(\xi_t) = -\mathbb{E}[x_t^1 \mid \xi_t] = K\xi, \qquad K = \frac{\mathrm{Var}(x_t^1)}{\mathrm{Var}(x_t^1) + \sigma_e^2} = \frac{\alpha^2\sigma_0^2}{\alpha^2\sigma_0^2 + \sigma_e^2}.$$

The loss from imperfect private sector misperception of x_1 is:

$$\mathbb{E}[\mathrm{Var}(x_t^1 \mid \xi)] = \mathrm{Var}(x_t^1)\big(1-K\big) = \frac{\alpha^2\sigma_0^2\,\sigma_e^2}{\alpha^2\sigma_0^2+\sigma_e^2}.$$

The adjustment cost is

$$\mathbb{E}[(x_t^1 - x_t^0)^2] = (\alpha-1)^2\sigma_0^2.$$

Thus

$$\mathcal{L}_{\text{lin}}(\alpha) = \frac{\alpha^2\sigma_0^2\,\sigma_e^2}{\alpha^2\sigma_0^2+\sigma_e^2} + \gamma\,(\alpha-1)^2\sigma_0^2, \qquad \mathcal{L}_{\text{lin}}^\star = \min_\alpha \mathcal{L}_{\text{lin}}(\alpha). \tag{11}$$

When γ is nontrivial, the optimizer may choose α substantially below one to limit policy adjustments. But shrinking α also shrinks the informativeness of $\xi = x_1 + e$, leaving the private sector unable to infer x_1 and offset x effectively.

4.4 Discrete policy rule

Allowing for a 1-bit quantized rule, where policy is set to bring the value of the intermediate output gap x_t^i to one of two levels $\{-m, m\}$:

$$\mathcal{F}_{Q1} = \{f(x_t^0) = -m\,\mathrm{sign}(x_t^0) :\ m \geq 0\}.$$

This rule uses only the sign of the fundamental, mapping the shock into one of two discrete policy stances $\pm m$.

In this section, we consider a discrete (or quantized) policy regime, where the central bank sets policy in order to achieve an intermediate output gap from among two discrete levels i.e $\{+m, -m\}$ conditional on the observed disturbance x_t^0. This represents a policy rule under 1-bit quantization, since the choice between the two discrete levels amounts to a single "bit" of information.

Under the 1-bit quantized rule $x_1 = -m\,\mathrm{sign}(x_0)$, the central bank deliberately chooses *two well-separated action levels*, making the signal $\xi = x_1 + e$ informative about the shock to the underlying fundamental x_1 even when the noise σ_e is large. The optimal decoder remains the posterior mean:

$$a^{\star PS}(\xi) = -\mathbb{E}[x_1 \mid \xi].$$

Because $x_1 \in \{-m, +m\}$ with equal prior probability (symmetry of x_0), Bayes' rule implies

$$\Pr(x_1 = +m \mid \xi) = \frac{1}{1+\exp\!\left(-\frac{2m\xi}{\sigma_e^2}\right)},$$

and therefore the optimal mapping from the noisy observation to PS action is given by:

$$a^\star(\xi) = -m\,(2\Pr(x_1 = +m \mid \xi) - 1) = -m\,\tanh\!\left(\frac{m\xi}{\sigma_e^2}\right). \tag{12}$$

The residual stabilization term becomes

$$\mathbb{E}\big[\mathrm{Var}(x_1 \mid \xi)\big] = \mathbb{E}\big[m^2 - a^\star(\xi)^2\big] = m^2\,\mathbb{E}\left[1 - \tanh^2\!\left(\frac{m\xi}{\sigma_e^2}\right)\right],$$

which falls rapidly when m is large relative to σ_e, because the two regimes become easy to distinguish.

The adjustment cost term can be computed in closed form. Since $x_t^1 = -m\,\mathrm{sign}(x_t^0)$,

$$\mathbb{E}[(x_t^1 - x_t^0)^2] = \mathbb{E}[x_0^2] + \mathbb{E}[x_1^2] - 2\mathbb{E}[x_1 x_0] = \sigma_0^2 + m^2 - 2m\,\mathbb{E}[|x_0|] = \sigma_0^2 + m^2 - 2m\,\sigma_0\sqrt{\frac{2}{\pi}}.$$

Hence the loss under this "discrete" policy regime is:

$$\mathcal{L}_{Q1}(m) = m^2\,\mathbb{E}\left[1 - \tanh^2\left(\frac{m\xi}{\sigma_e^2}\right)\right] + \gamma\left(\sigma_0^2 + m^2 - 2m\,\sigma_0\sqrt{\frac{2}{\pi}}\right), \qquad \mathcal{L}_{Q1}^\star = \min_{m\geq 0}\mathcal{L}_{Q1}(m). \tag{13}$$

The quantized rule can dominate the linear rule because mapping the fundamental shock x_t^0 to discrete levels in x_t^1 makes it easier to decode for the private sector agents. Increasing m makes ξ much more informative about the true output-gap x_t^1 (thereby reducing conditional variance $\mathbb{E}[\mathrm{Var}(x_1 \mid \xi)]$). In effect, the central bank prefers policy to be *coarsely* optimal, but also easy to decode.

4.5 Numerical dominance in a baseline calibration

In order to numerically illustrate the result, I use the following calibration:

$$\sigma_0 = 1, \qquad \sigma_e = 0.5, \qquad \gamma = 0.2. \qquad \tau = 1.0$$

A grid search over α and m yields: (i) linear optimum $\alpha^\star \approx 0.224$ with loss $\mathcal{L}_{\mathrm{lin}}^\star \approx 0.162$; and (ii) quantized optimum $m^\star \approx 1.16$ with loss $\mathcal{L}_{Q1}^\star \approx 0.141$. Thus

$$\frac{\mathcal{L}_{Q1}^\star}{\mathcal{L}_{\mathrm{lin}}^\star} \approx 0.87,$$

a welfare improvement of roughly 13%. The result is economically intuitive: because σ_e is non-trivial, the private sector's observations are noisy and cannot reliably infer small continuous variations in x_1 induced by a linear policy rule. Figure 1 plots the optimal policy under both linear and quantized policy regimes, while Figure 2 plots the corresponding optimal private-sector response.

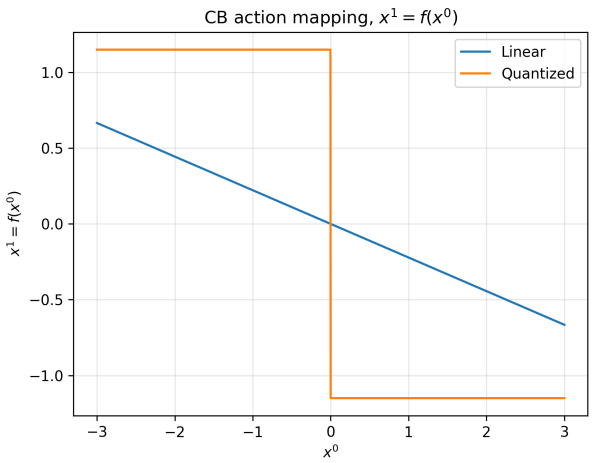


Figure 1: The Central Bank (CB)'s optimal policy response under the two policy classes

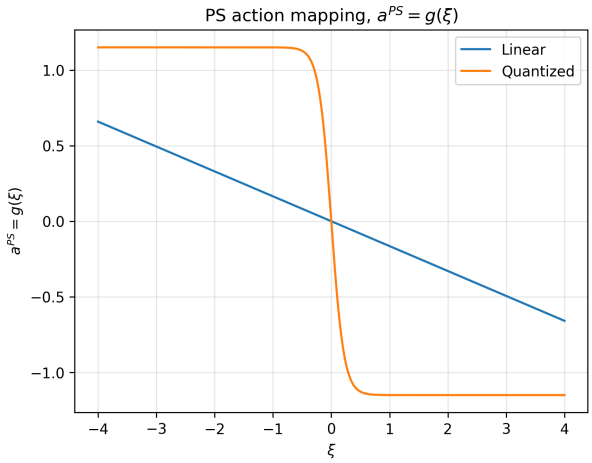


Figure 2: The Private Sector (PS)'s optimal policy response under the two policy classes

The quantized rule instead uses a discrete, *high-contrast* policy that is robustly inferable even through noise, enabling private agents to decode the fundamental reliably and the expectational channel to stabilize more effectively.

4.6 Discussion

Figure 3 shows how the relative gain from the nonlinear policy depends on the cost of interest-rate adjustments as well as the variance of the i.i.d noise in the private-sector's observations. The gain decreases with the increase in the magnitude of private-agents' observation noise, as signaling becomes less reliable. As the interest-rate adjustments become more costly, the relative gain from adopting the nonlinear policy rule increases.

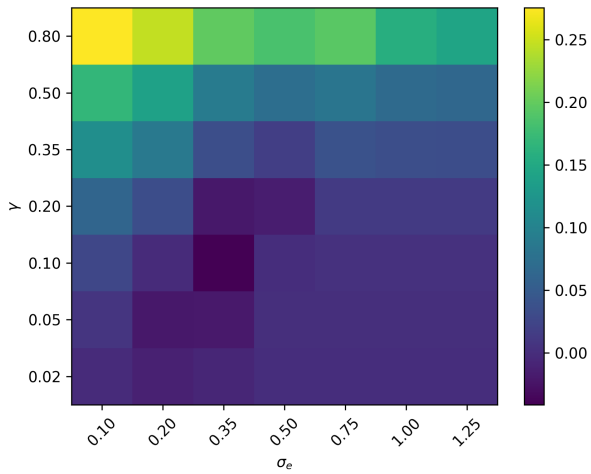


Figure 3: Relative improvement in social loss from quantized policy regime

Quantization is most valuable when: (i) the private sector's observation noise σ_e is material (communication friction); and (ii) the CB's adjustment cost λ is nontrivial discouraging stabilization through policy alone. In these environments, the CB may find it optimal to set policy so as to minimize the *knowledge gradient* rather than choosing to fully stabilize against the shock with linear feedback rules.

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A Inflation stabilization as part of social objective

So far, we have discussed social objectives that are independent of inflation. But adding an incentive to stabilize inflation to the social loss function leads to qualitatively similar results.

To see this, we can work with a larger model, with a reduced-form IS curve similar to the last section,

$$x = -\tau\big(i - r^n - \pi^e\big), \tag{14}$$

and adding a canonical Phillips curve

$$\pi \;=\; \kappa x + \beta \pi^e, \tag{15}$$

With a similar information structure: the central bank (CB) observes r^n and chooses $i = f(r^n)$; an intermediate variable

$$x_1 \;\equiv\; -\tau(i - r^n) \quad \Longleftrightarrow \quad i \;=\; r^n - \tau^{-1} x_1; \tag{16}$$

private agents observe $\xi = x_1 + e$ with $e \sim \mathcal{N}(0, \sigma_e^2)$ and choose $\pi^e = g(\xi)$.

Let the social objective be

$$L \;=\; x^2 + \chi\, \pi^2 + \lambda i^2, \qquad \chi \geq 0. \tag{17}$$

Define the rescaled fundamental $x_0 \equiv \tau r^n$ and the private-sector action

$$a(\xi) \;\equiv\; -\tau \pi^e(\xi). \tag{18}$$

Then minimizing $\mathbb{E}[L]$ over measurable policies (f, g) is equivalent to minimizing over measurable policies (f, g) the team objective

$$\mathcal{L}(f, g) \;=\; \mathbb{E}\Big[\begin{pmatrix} x_1 \\ a(\xi) \end{pmatrix}^{\top} Q(\chi) \begin{pmatrix} x_1 \\ a(\xi) \end{pmatrix} + \; \gamma\, (x_1 - x_0)^2 \Big], \tag{19}$$

subject to the nonclassical information structure $x_0 \to x_1 = f(x_0) \to \xi = x_1 + e \to a = g(\xi)$, where $\gamma \equiv \lambda \sigma^2$ and

$$Q(\chi) \;\equiv\; \begin{pmatrix} 1 + \chi \kappa^2 & -1 - \chi\, \kappa(\kappa + \beta \tau^{-1}) \\ -1 - \chi\, \kappa(\kappa + \beta \tau^{-1}) & 1 + \chi(\kappa + \beta \tau^{-1})^2 \end{pmatrix}. \tag{20}$$

The objective is still quadratic in (x_1, a) plus the control cost $\gamma(x_1 - x_0)^2$.

Suppress time subscripts. From (14) and (16),

$$x = -\tau(i - r^n - \pi^e) = -\tau(i - r^n) - \tau \pi^e = x_1 + \tau \pi^e = x_1 - a(\xi). \tag{21}$$

Rewrite inflation using (15) and (18):

$$\begin{aligned} \pi &= \kappa(x_1 - a(\xi)) + \beta\, \pi^e = \kappa_x (x_1 - a(\xi)) + \beta(-\tau^{-1} a(\xi)) \\ &= \kappa x_1 - (\kappa + \beta \tau^{-1})\, a(\xi). \end{aligned} \tag{22}$$

Finally, the interest-rate term becomes a quadratic cost in $(x_1 - x_0)$. Using $x_0 \equiv \tau r^n$ so that $r^n = \tau^{-1} x_0$, (16) implies

$$\lambda i^2 = \lambda (r^n - \tau^{-1} x_1)^2 = \frac{\lambda}{\tau^2} (x_0 - x_1)^2 = \gamma (x_1 - x_0)^2, \qquad \gamma \equiv \frac{\lambda}{\tau^2}. \tag{23}$$

Substituting (21), (22), and (23) into $L = x^2 + \chi \pi^2 + \lambda i^2$ yields

$$\begin{aligned} L &= (x_1 - a)^2 + \chi \big(\kappa x_1 - (\kappa + \beta \tau^{-1}) a\big)^2 + \gamma (x_1 - x_0)^2 \\ &= \begin{pmatrix} x_1 \\ a \end{pmatrix}^{\top} Q(\chi) \begin{pmatrix} x_1 \\ a \end{pmatrix} + \gamma (x_1 - x_0)^2, \end{aligned} \tag{24}$$

where collecting coefficients gives (20). Taking expectations under the information structure $x_0 \to x_1 \to \xi \to a$ yields (19). The information structure is nonclassical because the CB's choice x_1 enters the private sector's observation equation $\xi = x_1 + e$, so Controller 1's action shapes Controller 2's posterior and response.

If u is mean-zero and independent of (x_0, e), the additional linear term in (19) vanishes in expectation, and u contributes only an additive constant $\chi\mathbb{E}[u^2]$.